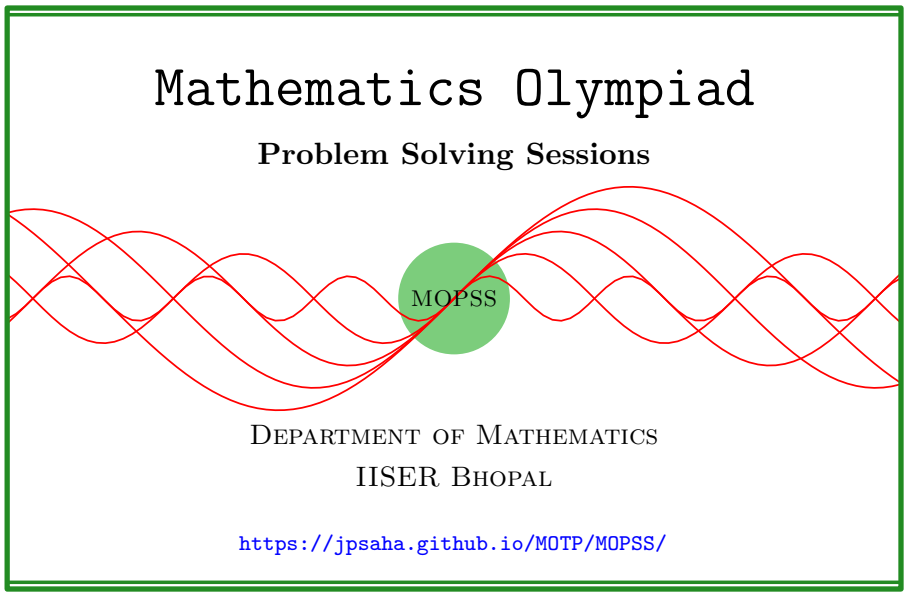
# Pythagoras' theorem

MOPSS




## Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

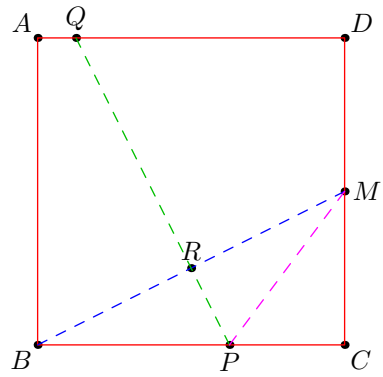



Figure 1: RMO 1990 P3, Exercise 1

## List of problems and examples

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## §1 Pythagoras' theorem

**Exercise 1** (RMO 1990 P3, ). A square sheet of paper $ABCD$ is so folded that $B$ falls on the mid-point $M$ of $CD$. Prove that the crease will divide $BC$ in the ratio $5 : 3$.

**Solution 1.** Let the crease intersect $BC$ at $P$, and $AD$ at $Q$. Let $R$ denote the midpoint of $BM$. Then $PQ$ is the perpendicular bisector of $BM$. Note that $\triangle BPR$ and $\triangle MPR$ are congruent. This gives $BP = PM$. Applying the Pythagoras' theorem to $\triangle CPM$, we obtain

$$CM^2 + CP^2 = PM^2,$$

which gives

$$CM^2 + CP^2 = BP^2.$$

This implies that

$$CM^2 = (BP - CP)(BP + CP),$$

which yields

$$BP - CP = \frac{CM^2}{BP + CP} = \frac{BC}{4} = \frac{BP + CP}{4}.$$

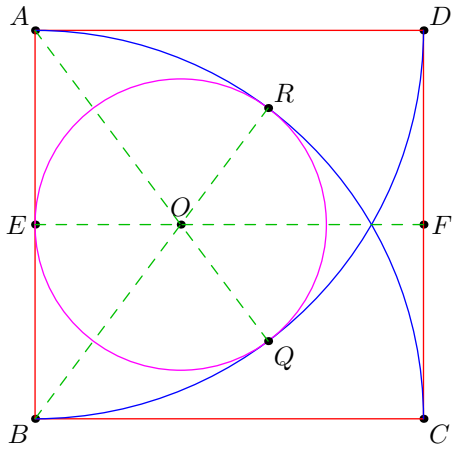



Figure 2: RMO 2012b P1, Exercise 2

We get

$$\frac{BP}{CP} = \frac{1+4}{4-1} = \frac{5}{3}.$$

■

**Exercise 2** (RMO 2012b P1, ). Let $ABCD$ be a unit square. Draw a quadrant of a circle with $A$ as centre and $B, D$ as endpoints of the arc. Similarly, draw a quadrant of a circle with $B$ as centre and $A, C$ as endpoints of the arc. Inscribe a circle $\Gamma$ touching the arc $AC$ internally, the arc $BD$ internally and also touching the side $AB$. Find the radius of the circle $\Gamma$.

**Solution 2.** Let $E$ denote the midpoint of $AB$, and $F$ denote the midpoint of $CD$. Note that $EF$ is parallel to $AB$ and $CD$, and has length equal to that of $AB$ and $CD$. Indeed, the line passing through the midpoint $E$ of $AB$ and parallel to $BC$ bisects $AC$. Since this line is also parallel to $AD$, it bisects $CD$, hence passes through $F$. Thus, $EF$ is parallel to $AB$ and $CD$, and has length equal to that of $AB$ and $CD$.

Note that $\triangle ADF, \triangle BCF$ are congruent, and hence $AF = BF$. This implies that $\triangle AEF, \triangle BEF$ are congruent, and hence $EF$ is a perpendicular bisector of $AB$.

Let $O$ denote the centre of the circle $\Gamma$. Let $Q$ denote the point of tangency of $\Gamma$ with the arc $BD$, and $R$ denote the point of tangency of $\Gamma$ with the arc $AC$. Since $\Gamma$ and the arc $BD$ touch at $Q$, it follows that $AQ, QO$ are perpendicular to their common tangent at $Q$. Hence, the points $A, Q, O$ are collinear, and this gives

$$AO = AQ - OQ.$$

Similarly, since $\Gamma$ and the arc $AC$ touch at $R$, it follows that $B, R, O$ are collinear, and this gives

$$BO = BR - OR.$$

This implies that

$$BR - OR = BC - OR = AD - OR = AQ - OQ = AO.$$

Using an argument as above, it turns out that $\triangle AEO, \triangle BEO$ are congruent, and hence $EO$ is a perpendicular bisector of $AB$. This implies that $O$ lies on the line $EF$.

Since $\Gamma$ is a circle touching $AB$, it follows that the point of tangency of $\Gamma$ with $AB$ is the foot of the perpendicular from $O$ to $AB$. Since $OE$ is perpendicular to $AB$, it follows that the point of tangency of $\Gamma$ with $AB$ is $E$. Thus, the radius of $\Gamma$ is equal to $OE$.

Let $r$ denote the radius of $\Gamma$, and $s$ denote the length of the sides of $ABCD$. It follows from the above that

$$EO = r, AO = AQ - OQ = s - r.$$

Applying the Pythagoras' theorem to $\triangle AEO$, we obtain

$$r^2 + \left(\frac{s}{2}\right)^2 = (s - r)^2,$$

which gives $r = \frac{3s}{8} = \frac{3}{8}$. ■

**Exercise 3** (RMO 2012c P1, ). Let $ABCD$ be a unit square. Draw a quadrant of a circle with $A$ as centre and $B, D$ as endpoints of the arc. Similarly, draw a quadrant of a circle with $B$ as centre and $A, C$ as endpoints of the arc. Inscribe a circle $\Gamma$ touching the arcs $AC$ and $BD$ both externally and also touching the side $CD$. Find the radius of the circle $\Gamma$.

**Solution 3.** Let $E$ denote the midpoint of $AB$, and $F$ denote the midpoint of $CD$. Note that $EF$ is parallel to $AB$ and $CD$, and has length equal to that of $AB$ and $CD$. Indeed, the line passing through the midpoint $E$ of $AB$ and parallel to $BC$ bisects $AC$. Since this line is also parallel to $AD$, it bisects $CD$, hence passes through $F$. Thus, $EF$ is parallel to $AB$ and $CD$, and has length equal to that of $AB$ and $CD$.

Note that $\triangle ADF, \triangle BCF$ are congruent, and hence $AF = BF$. This implies that $\triangle AEF, \triangle BEF$ are congruent, and hence $EF$ is a perpendicular bisector of $AB$. Similarly, it follows that $EF$ is a perpendicular bisector of $CD$.

Let $O$ denote the centre of the circle $\Gamma$. Let $Q$ denote the point of tangency of $\Gamma$ with the arc $BD$, and $R$ denote the point of tangency of $\Gamma$ with the arc $AC$. Since $\Gamma$ and the arc $BD$ touch at $Q$, it follows that $AQ, QO$ are perpendicular to their common tangent at $Q$. Hence, the points $A, Q, O$ are collinear, and this gives

$$AO = AQ + QO.$$

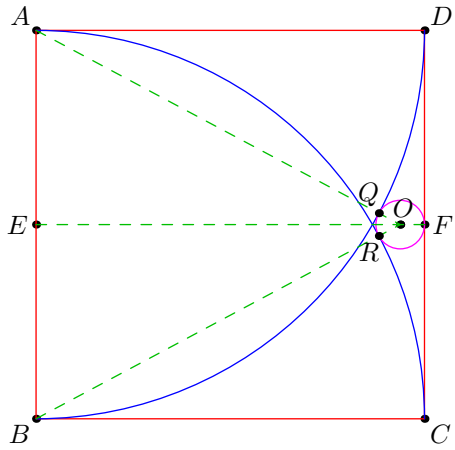



Figure 3: RMO 2012c P1, Exercise 3

Similarly, since $\Gamma$ and the arc $AC$ touch at $R$, it follows that $BR, RO$ are perpendicular to their common tangent at $R$. Hence, the points $B, R, O$ are collinear, and this gives

$$BO = BR + RO.$$

This implies that

$$BO = BR + RO = BC + QO = AD + QO = AQ + QO = AO.$$

Using an argument as above, it turns out that $\triangle AEO, \triangle BEO$ are congruent, and hence $EO$ is a perpendicular bisector of $AB$. This implies that $O$ lies on the line $EF$.

Since $\Gamma$ is a circle touching $CD$, it follows that the point of tangency of $\Gamma$ with $CD$ is the foot of the perpendicular from $O$ to $CD$. Since $OF$ is perpendicular to $CD$, it follows that the point of tangency of $\Gamma$ with $CD$ is $F$. Thus, the radius of $\Gamma$ is equal to $OF$. This gives

$$EO = EF - OF.$$

Let $r$ denote the radius of $\Gamma$, and $s$ denote the length of the sides of $ABCD$. It follows from the above that

$$EO = EF - OF = s - r, AO = AQ + QO = s + r.$$

Applying the Pythagoras' theorem to $\triangle AEO$, we obtain

$$(s - r)^2 + \left(\frac{s}{2}\right)^2 = (s + r)^2,$$

which gives $r = \frac{s}{16} = \frac{1}{16}$. ■

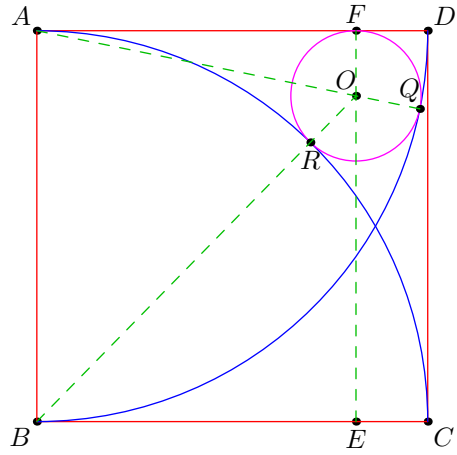



Figure 4: RMO 2012d P1, Exercise 4

**Exercise 4** (RMO 2012d P1, ). Let $ABCD$ be a unit square. Draw a quadrant of a circle with $A$ as centre and $B, D$ as endpoints of the arc. Similarly, draw a quadrant of a circle with $B$ as centre and $A, C$ as endpoints of the arc. Inscribe a circle $\Gamma$ touching the arc $AC$ externally, the arc $BD$ internally and also touching the side $AD$. Find the radius of the circle $\Gamma$.

**Solution 4.** Let $O$ denote the centre of the circle $\Gamma$, and $F$ denote the point of tangency of $\Gamma$ with the side $AD$. Note that $OF$ is perpendicular to $AD$, and hence $OF$ is parallel to $AB$. Let $E$ denote the foot of the perpendicular from $O$ to $BC$. Note that $OE$ is parallel to $AB$. It follows that $E, O, F$ are collinear, and hence

$$EF = EO + OF.$$

Since $EF, AB$ are parallel and so are $E, AF$, it follows that $ABEF$ is a parallelogram, and hence, we obtain $EF = AB, BE = AF$.

Let $O$ denote the centre of the circle $\Gamma$. Let $Q$ denote the point of tangency of $\Gamma$ with the arc $BD$, and $R$ denote the point of tangency of $\Gamma$ with the arc $AC$. Since $\Gamma$ and the arc $BD$ touch at $Q$, it follows that $AQ, QO$ are perpendicular to their common tangent at $Q$. Hence, the points $A, Q, O$ are collinear, and this gives

$$AO = AQ - OQ.$$

Similarly, since $\Gamma$ and the arc $AC$ touch at $R$, it follows that $BR, RO$ are perpendicular to their common tangent at $R$. Hence, the points $B, R, O$ are collinear, and this gives

$$BO = BR + RO.$$

Let $r$ denote the radius of $\Gamma$, and $s$ denote the length of the sides of $ABCD$. It follows from the above that

$$BO = BR + RO = s + r, AO = AQ - OQ = s - r.$$

Applying the Pythagoras' theorem to $\triangle AOF, \triangle BEO$, we obtain

$$BE^2 + OE^2 = (s+r)^2, AF^2 + OF^2 = (s-r)^2.$$

Using $BE = AF$, we get

$$(OE - OF)(OE + OF) = 4sr,$$

which yields

$$(EF - 2 \cdot OF)(EF) = 4sr.$$

This gives

$$(s - 2r)s = 4sr,$$

implying that

$$r = \frac{s}{6} = \frac{1}{6}.$$

■

**Exercise 5** (IOQM 2024 P15, ). Let $X$ be the set consisting of twenty positive integers $n, n+2, \ldots, n+38$. The smallest value of $n$ for which any three numbers $a, b, c \in X$, not necessarily distinct, form the sides of an acute-angled triangle is:

**Summary** — Use Pythagoras' theorem.

**Walkthrough** —

**(a)** Use the fact that[a] if $a, b, c$ are the sides of an acute-angled triangle with $c$ denoting the largest side, then $a^2 + b^2 > c^2$ holds.

**(b)** Show that for any such $n$, the inequality

$$(n+38)^2 < 2n^2$$

holds, which yields $n \geq 92$.

**(c)** Prove that for $n = 92$, the inequality

$$2(n+2i)^2 > (n+2i+2)^2$$

holds for $i = 0, 1, \ldots, 18$.

**(d)** Note that if suffices to show that

$$2m^2 > (m+2)^2$$

for any $m \in \{92, 94, \ldots, 128\}$.

What is the role of part (c)? Does it guarantee that for $n = 92$, the set $X$ has the stated property?

[a] How to prove it? Does the **converse** of this statement hold? In other words, if $a, b, c$ are positive real numbers satisfying $a^2 + b^2 > c^2$, then does it follow that $a, b, c$ are the sides of an acute-angled triangle? Even if it doesn't, would it be true under additional hypothesis?

**Solution 5.** ■

## References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)